

Digital Image Processing

Digital Camera

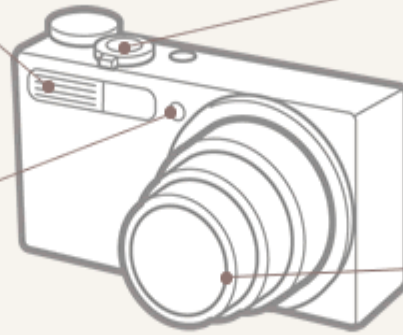
- ▶ A digital camera works by capturing light with a lens, which focuses it onto an image sensor that converts the light into electrical signals. These signals are then processed by a CPU to create a digital image, which is stored on a memory card. Key components include the lens, aperture, shutter, image sensor, processor, and storage

Flash

The flash fires to give an instantaneous burst of light to supplement image brightness, such as when photographing in low-light situations.

AF auxiliary light

The AF auxiliary light fires to help the autofocus (AF) function properly focus the lens in low-light situations.



Shutter release button

This button works in two steps. When you press it lightly (=half-press), the camera focuses using the autofocus, and when you press it all the way down (=full-press), the camera releases the shutter to take the photograph.

Lens

With the lens playing a role comparable to that of the human eye, the camera takes the photograph using the light that enters through the lens. There are fixed focal length lenses and zoom lenses.

Zoom lever

When a camera has a zoom lens, you can adjust the range of the scene that appears in the picture, thereby shooting a wide expanse or magnifying a smaller

Components Inside a Digital Camera

- ▶ **1.Lens:** Focuses light onto the image sensor.
- ▶ **2.Image Sensor (CCD/CMOS):** Converts light into electrical signals (analog).
- ▶ **3. Analog-to-Digital Converter (ADC):** Converts analog signals to digital data.
- ▶ **4.Image Processing Unit (IPU):** Processes digital data into images (demosaicing, white balancing, noise reduction).
- ▶ **5. Memory Buffer:** Temporarily stores processed images.
- ▶ **6. Storage Media (Memory Card):** Stores captured images.
- ▶ **7. Digital Camera Processor:** Controls camera functions and image processing.

Working of a Digital Camera

➤ **Image Acquisition**

- Light from the scene enters the camera. Lens focuses the light onto the image sensor.

➤ **Optical System Control**

- Aperture controls how much light enters. Shutter controls how long light falls on the sensor.
- This determines exposure, which affects brightness and clarity.

➤ **Image Sensing**

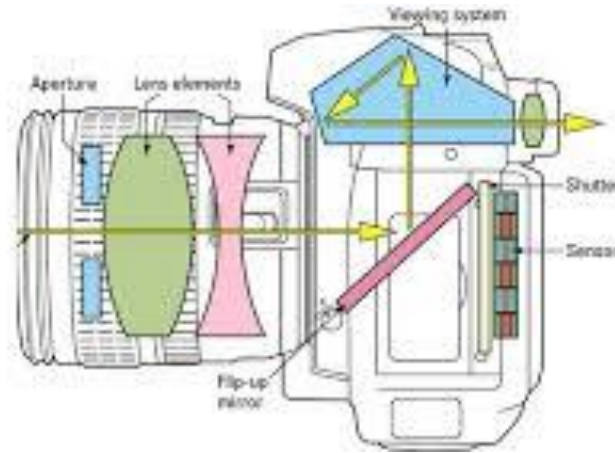
- Sensor = CCD or CMOS It converts focused light into electrical charges. Each sensor element = pixel → More light = larger electrical signal → Less light = smaller signal
- This creates analog image data.

Working of a Digital Camera

- **Analog-to-Digital Conversion (ADC)**
- Sensor output is analog. ADC converts it into digital numbers (0s and 1s). DIP works only on digital data, so this step is essential.
- **Image Signal Processing (ISP)**
- Digital image is processed internally: Noise reduction Color interpolation (Bayer filter processing) White balance Sharpening Contrast adjustment Compression (JPEG, PNG) Finally, the data becomes a digital image file.
- **Image Storage**
- Processed image is saved to memory card.

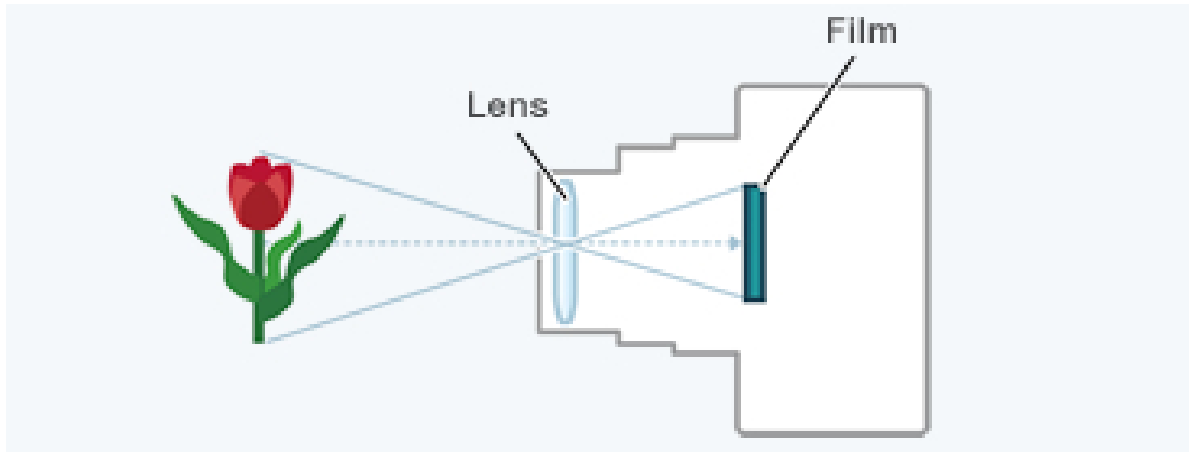
Digital Image Processing Steps

- **1. Image Acquisition:** Captures light and converts it to digital data.
- **2. Pre-processing:** Adjusts image quality (noise reduction, sharpening).
- **3. Demosaicing:** Reconstructs full-color images from raw data.
- **4. Post-processing:** Enhances image quality (color correction, compression).
- **5. Image Storage:** Saves processed images to storage media. These components and steps work together to capture and process digital images.



Digital Camera photographic mechanism

- ▶ **Film camera**
- ▶ With a film camera, the light entering through the lens strikes the film. Developing the film makes it possible to see the photograph that was taken.



Digital Camera photographic mechanism

- ▶ **Digital camera**
- ▶ With a digital camera, the light entering through the lens strikes an image sensor. The signal output by the image sensor is processed within the camera to create image data, which is stored on the memory card. The image can be simultaneously viewed on the picture display

